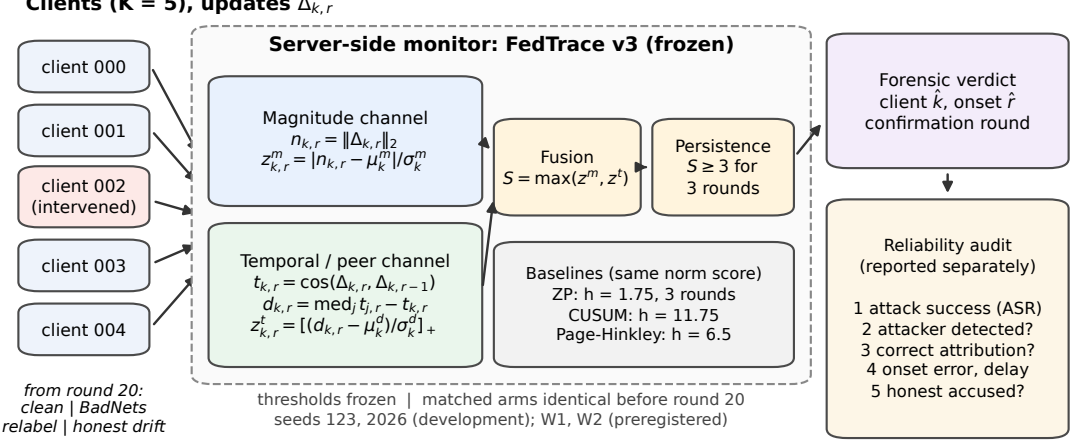
Clients (K = 5), updates $\Delta_{k,r}$
client 000
client 001
client 002 (intervened)
client 003
client 004
from round 20: clean | BadNets relabel | honest drift
Server-side monitor: FedTrace v3 (frozen)
Magnitude channel
$n_{k,r} = \|\Delta_{k,r}\|_2$
$z^m_{k,r} = |n_{k,r} - \mu^m_k| / \sigma^m_k$
Temporal / peer channel
$t_{k,r} = \cos(\Delta_{k,r}, \Delta_{k,r-1})$
$d_{k,r} = \mathrm{med}_j\, t_{j,r} - t_{k,r}$
$z^t_{k,r} = [(d_{k,r} - \mu^d_k)/\sigma^d_k]_+$
Fusion
$S = \max(z^m, z^t)$
Persistence
$S \geq 3$ for 3 rounds
Baselines (same norm score)
ZP: h = 1.75, 3 rounds
CUSUM: h = 11.75
Page-Hinkley: h = 6.5
thresholds frozen | matched arms identical before round 20
seeds 123, 2026 (development); W1, W2 (preregistered)
Forensic verdict
client $\hat{k}$, onset $\hat{r}$
confirmation round
Reliability audit
(reported separately)
1 attack success (ASR)
2 attacker detected?
3 correct attribution?
4 onset error, delay
5 honest accused?